

3.8 Polymerisation

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.8 Polymerisation
Difficulty	Medium

Time allowed: 20

Score: /10

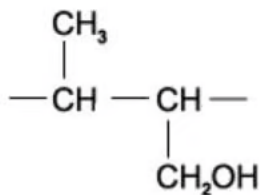
Percentage: /100

Question 1

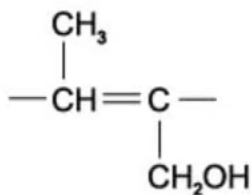
Plasticisers, resins and many other chemicals can be made synthetically by polymerisation of a variety of monomers including but-2-en-1-ol, $\text{CH}_3\text{CH}=\text{CHCH}_2\text{OH}$.

Which structure is the repeating unit in poly(but-2-en-1-ol)?

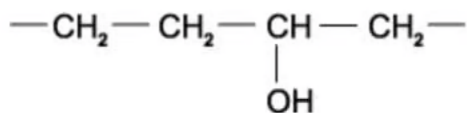
A.



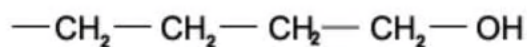
B.



C.



D.



[1 mark]

Question 2

Which statement about PVC is **incorrect**?

- A. PVC molecules are saturated
- B. The repeat unit of PVC is $-(CHC/CHC)-$
- C. Combustion of PVC waste produces a highly acidic gas
- D. The empirical formula of PVC is the same as the empirical formula of its monomer

[1 mark]

Question 3

PVC is produced through the polymerisation of chloroethene.

How does the carbon-carbon bond in chloroethene compare with that in PVC?

- A. Longer and stronger
- B. Longer and weaker
- C. Shorter and stronger
- D. Shorter and weaker

[1 mark]

Question 4

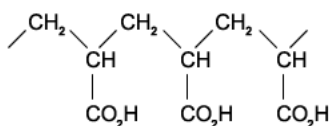
What are the C-C-C bond angles along the polymeric chain in PVC?

- A. Half are 109.5° and half are 120°
- B. They are all 109.5°
- C. They are all 120°
- D. They are all 180°

[1 mark]

Question 5

The absorbent material in babies' disposable nappies is made from the addition polymer shown.



From which monomer could this addition polymer be obtained?

- A. $\text{HOCH}_2\text{CH}_2\text{CO}_2\text{H}$
- B. $\text{HO}_2\text{CCH}=\text{CHCO}_2\text{H}$
- C. $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$
- D. $\text{H}_2\text{C}=\text{CHCO}_2\text{H}$

[1 mark]

Question 6

In the addition polymerisation of alkenes, which types of bond breakage and bond formation occur?

	Bond breakage	Bond formation
A	σ and π	σ only
B	σ and π	σ and π
C	π only	σ only
D	π only	σ and π

[1 mark]

Question 7

Cling film is made through the polymerisation of 1,1-dichloroethene. The product is dense with a high melting point and does not allow gases to pass through which makes it ideal for preserving food.

Which sequence appears in a short length of the polymer chain?

- A. $-\text{CH}_2\text{CCl}_2\text{CHClCHClCH}_2\text{CCl}_2-$
- B. $-\text{CCl}_2\text{CCl}_2\text{CCl}_2\text{CCl}_2\text{CCl}_2\text{CCl}_2-$
- C. $-\text{CHClCHClCHClCHClCHClCHCl}-$
- D. $-\text{CH}_2\text{CCl}_2\text{CH}_2\text{CCl}_2\text{CH}_2\text{CCl}_2-$

[1 mark]

Question 8

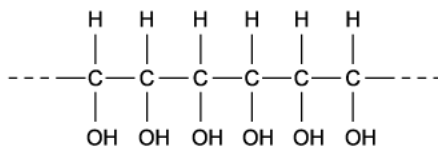
Which polymer is not made when 1-hydroxyethene and propene undergo additional polymerisation?

- A. $-\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{CH}_3)-$
- B. $-\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{CH}_3)-$
- C. $-\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CHCH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CHCH}_3-$
- D. $-\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}(\text{OH})-$

[1 mark]

Question 9

The diagram shows a short length of an addition polymer chain.



The polymer has a relative molecular mass of approximately 10 000.

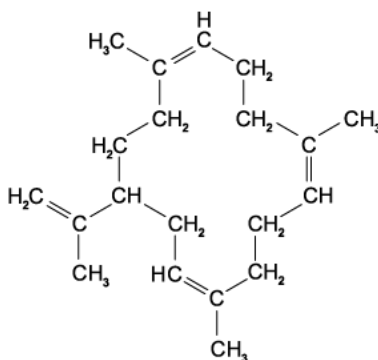
Approximately how many monomer units are joined together in each polymer molecule?

- A. 80
- B. 160
- C. 240
- D. 320

[1 mark]

Question 10

The addition of four identical monomer molecules produces the naturally occurring molecule shown below.



What could be the structural formula of the monomer?

- A. $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$
- B. $\text{CH}_3\text{CH}=\text{CHCH}=\text{CH}_2$
- C. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$
- D. $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}=\text{CH}_2$

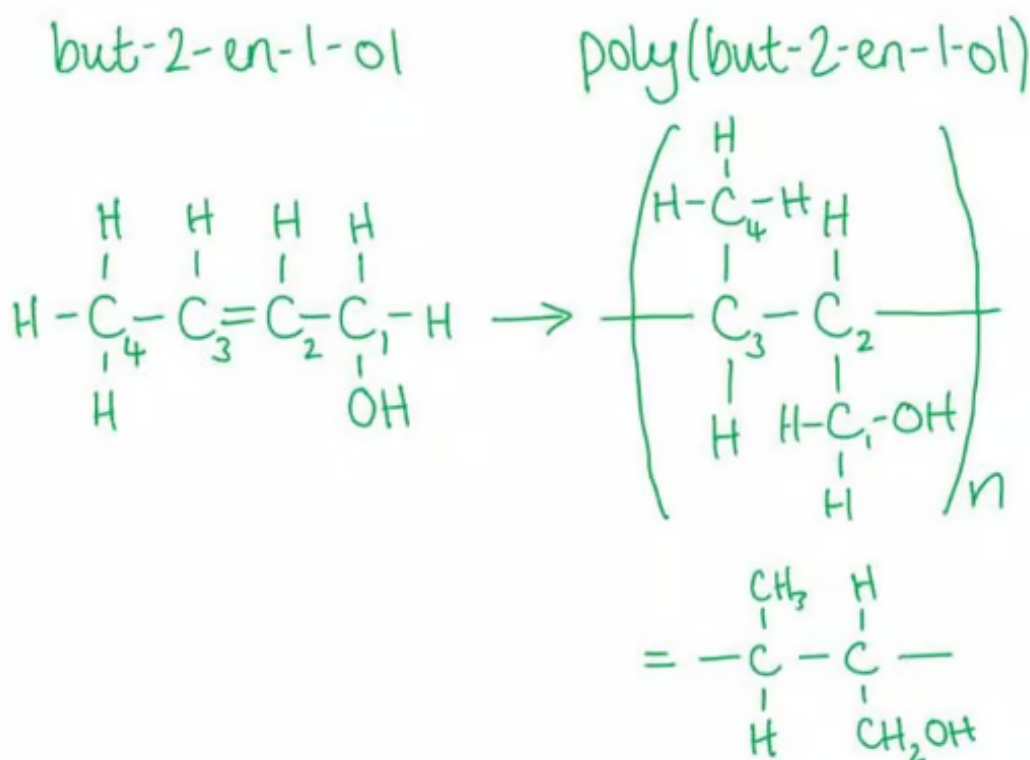
[1 mark]

Model Answers: Medium

1

The correct answer is **A** because:

- The chemical name but-2-en-1-ol tells us that each molecule is 4 carbons long (but), has a double bond between the 2nd and 3rd carbons (2-en) and an OH group attached to the 1st carbon (1-ol).
- A repeating unit is the monomer, in this case but-2-en-1-ol, but the double bond is broken to form one carbon carbon single bond and an electron for each carbon to form a new bond with the next monomer.



B is incorrect as repeating units do **not** contain **double** bonds.

C is incorrect as repeating units only have the **two carbons** involved in the double bond in the main chain, all other groups are attached to one of these carbons.

D is incorrect as repeating units only have the **two carbons** involved in the double bond

in the main chain, all other groups are attached to one of these carbons. In **addition**, **polymers** separate monomers are connected by carbon carbon single bonds **only**.

2

The correct answer is **B** because:

- PVC stands for polyvinylchloride, also known as poly(chloroethene).
- PVC is a very common polymer made from the addition polymerisation of **chloroethene, $\text{H}_2\text{C}=\text{CHCl}$**
- A repeating unit is the monomer, in this case chloroethene, but the double bond is broken to form one carbon carbon single bond and an electron for each carbon to form a new bond with the next monomer therefore the repeating unit is: – **(CH_2CHCl) –**

A is incorrect as the statement ‘PVC molecules are **saturated**’ is true, saturated means that there are **only single bonds**.

C is incorrect as the statement ‘combustion of PVC waste does produce a **highly acidic gas**’ is true, PVC combustion produces **hydrogen chloride** gas.

D is incorrect as the statement ‘the **empirical formula** of PVC is the **same** as the empirical formula of its monomer’ is true because this is the case for **all** addition polymers.

3

The correct answer is **C** because:

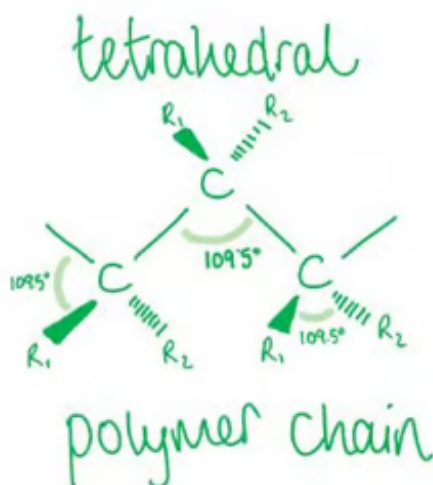
- During the polymerisation of chloroethene to PVC the carbon carbon **double bond breaks** to become a single bond and frees an electron per carbon to form new single bonds with adjacent monomers.
- Therefore, this question is asking the difference between a double carbon carbon bond in chloroethene and a single carbon carbon bond in PVC.
- **Double bonds** are **stronger** than single bonds because **more electrons** are being shared between the two carbons and therefore the carbons are held together tighter.

- **Stronger** bonds are **shorter** bonds as the strength of a bond pulls the two atoms closer together.
- Therefore, the carbon-carbon bond of chloroethene is **shorter and stronger** than in PVC.

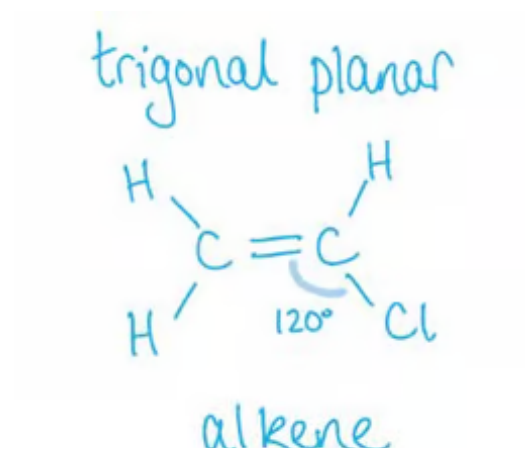
4

The correct answer is **B** because:

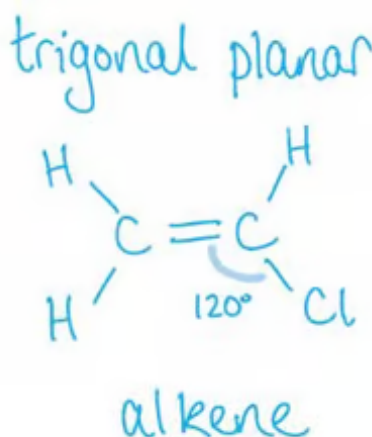
- The C-C-C bonds in the question refers to the carbon main chain, or 'backbone' of the PVC (polychloroethene) polymer.
- PVC is **fully saturated**, meaning there are only single bonds in the molecule.
- Carbons each make **4 bonds**, and these are in a **tetrahedral shape** to enable **maximum separation** between the electron pairs.
- **All angles are 109.5°** in tetrahedral molecules.



A & C are incorrect as they suggest that PVC the molecule contains a **trigonal planar** shape (which has **120°** angles), this shape is seen in **alkenes** such as the PVC monomer chloroethene.



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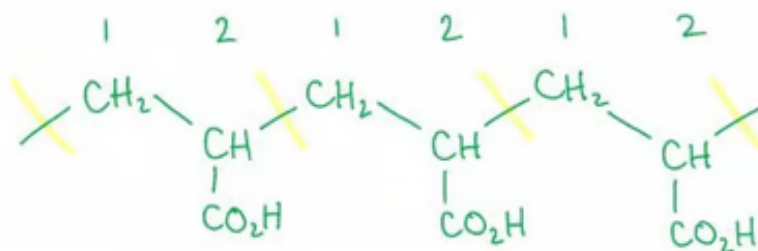


D is incorrect as they suggest that PVC the molecule contains a **linear** shape (which has **180°** angles), this shape is seen in carbon dioxide.

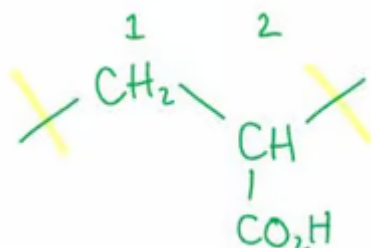
5

The correct answer is **D** because:

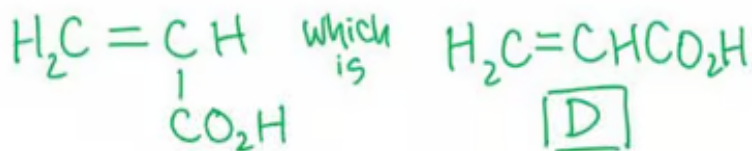
- Repeating units have 2 carbons in the polymer main chain.



- One **repeating unit**:

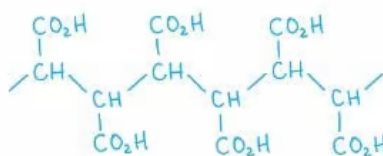


- Distinguishing a repeating unit leads us to the name of the polymer shown – **poly(prop-2-enoic acid)**, this means the monomer is prop-2-enoic acid.
- The **monomer** of addition polymers is a repeating unit with a **double bond**.



A & C are incorrect as these do not have double bonds and also do not have the correct functional groups – there are no hydroxy groups in this polymer.

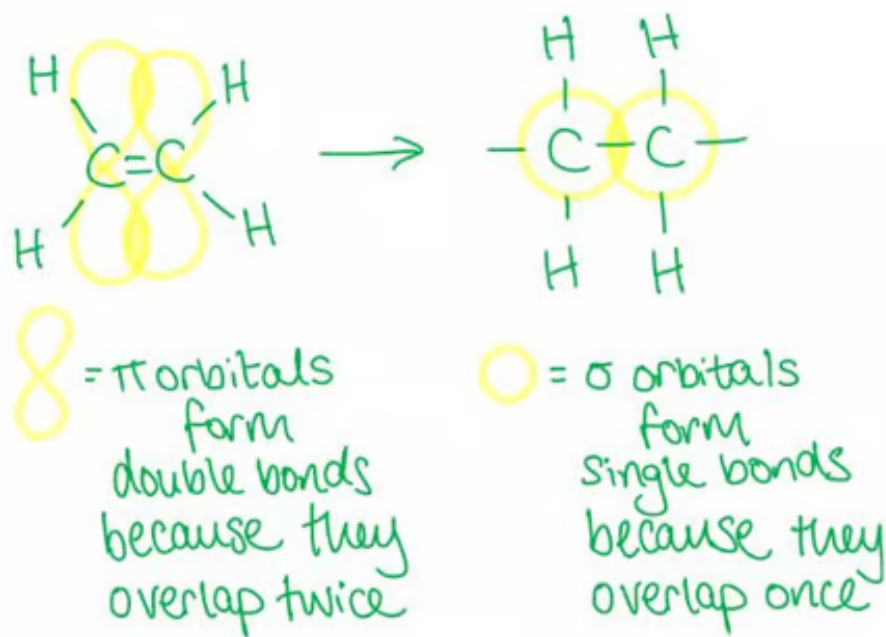
B is incorrect as this is the monomer for the polymer below: poly(prop-2-en-1,4-dioic acid)



6

The correct answer is **C** because:

- During addition polymerisation the double bond on the alkene monomer is broken to produce a single carbon-carbon bond within the monomer and frees one electron per carbon to form new single bonds with adjacent monomers.



- Therefore, a double ($\pi = \pi$) bond is broken and single ($\sigma = \sigma$) bond is formed.

A is incorrect as no sigma bonds are broken.

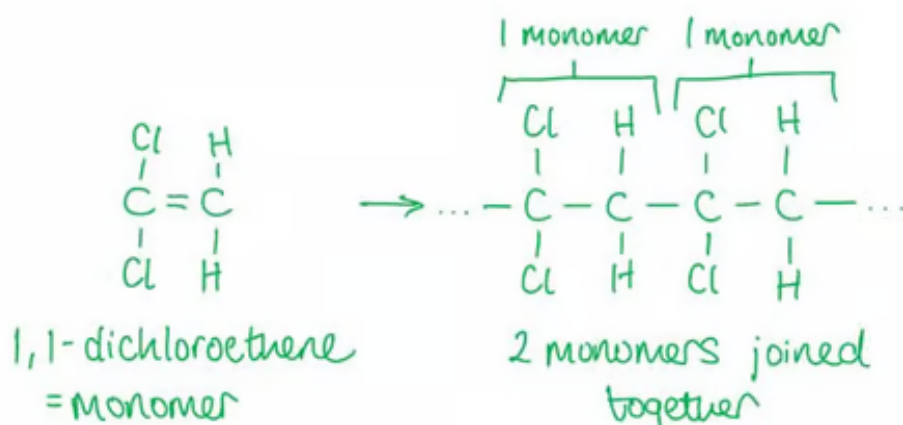
B is incorrect as no sigma bonds are broken, and no pi bonds are formed.

D is incorrect as no pi bonds are formed.

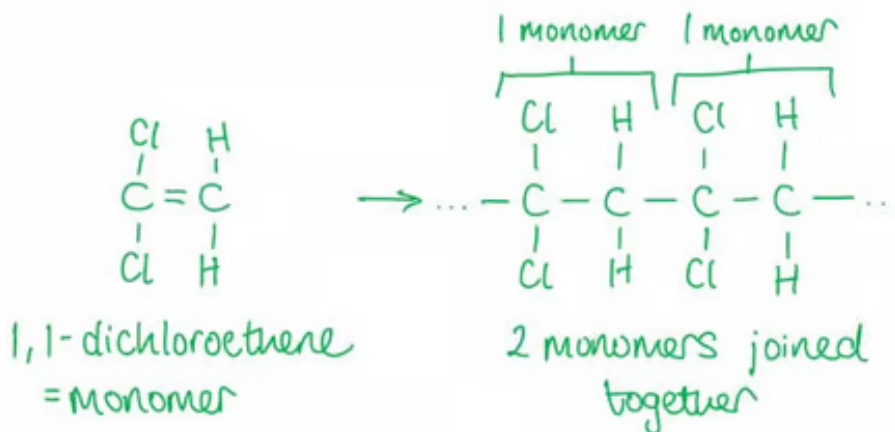
7

The correct answer is **D** because:

- 1,1-dichloroethene is a 2-carbon long alkene 'ethene' with 2 chlorine functional groups attached to the first carbon.
- In the case of ethene, the first carbon can be classed as either of the carbons because the molecule is symmetrical.
- The monomer is therefore CCl_2CH_2 or **CH_2CCl_2** .
- The polymer is a repeat of this as seen in option D.



- The polymer is a repeat of this as seen in option D.



A is incorrect as $-\text{[CH}_2\text{CCl}_2\text{CHC/CHC/CH}_2\text{CCl}_2\text{]}-$ the part in bold is incorrect because there is only one chlorine functional group attached to the carbon, not 2.

B is incorrect as $-\text{[CCl}_2\text{CCl}_2\text{CCl}_2\text{CCl}_2\text{CCl}_2\text{CCl}_2\text{]}-$ suggests there are **2 chlorine** functional groups **per carbon** which would make the monomer **1,1,2,2-tetrachloroethene**.

C is incorrect as $-\text{[CHC/CHC/CHC/CHC/CHC/CHC/]}-$ suggests there is one **chlorine** functional group **per carbon**, rather than both on the first carbon, which would make the monomer **1,2-dichloroethene**.

8

The correct answer is **C** because:

- An addition polymer made from two different monomers is called a **copolymer**.
- The monomers can be joined in a variety of manners, however **each monomer** will only have **2 carbons** joined within the main polymer chain.
- All other groups must be attached to one of the two main carbons and, when written in the format of this question, must be written **in brackets** – if not they are assumed to be part of the main chain.
- *Hydrogen atoms can be written with their carbon.*

- 4 x Hydrogen (1 g/mol)
- **Total molecular mass of a monomer:**
 - $(2 \times 12) + (2 \times 16) + (4 \times 1) = 60 \text{ g/mol}$
- **Number of monomers in the polymer (polymer $M_r \div$ monomer M_r):**
 - $10\,000 \div 60 = 166.67$
- The question uses '**approximate**' values, therefore **option B – 160** – is the approximate number of monomer units.

10

The correct answer is **D** because:

- The polymer molecule contains double bonds in its main chain which means that the monomer must have 2 sets of double bonds, which narrows the answer down to **B** or **D**.
- The 2 sets of double bonds 'open up' / break which causes a new double bond to be made in the middle of the monomer as well as single bonds to bond to adjacent monomers.
- To distinguish between the two, you can see that option D has a methyl group which is seen coming off the polymer chain.

